Cyclotomic Polynomials

Question 1

Write the polynomial $\Phi_{15}(X)$ in the form $\sum_{i=0}^{N} c_i X^i$.

Solution. We know that $X^{15}-1 = \Phi_1(X)\Phi_3(X)\Phi_5(X)\Phi_{15}(X)$. Further, we know that $X^5-1 = \Phi_1(X)\Phi_5(X)$, $X^3-1 = \Phi_1(X)\Phi_3(X)$ and $X-1 = \Phi_1(X)$.

Thus,

$$\Phi_{15}(X) = \frac{X^{15}-1}{X^5-1} \cdot \frac{X-1}{X^3-1}$$

By the usual geometric series, we have

$$\frac{X^{15}-1}{X^5-1} = (X^5)^2 + X^5 + 1$$
$$\frac{X^3-1}{X-1} = X^2 + X + 1$$

$$\begin{aligned} &X^{10} + X^5 + 1 = \\ &\quad X^7(X^3-1) + X^4(X^3-1) + X(X^3-1) + X^2(X^3-1) + X^2 + X + 1 \\ &\quad = ((X^8 - X^7) + (X^5 - X^4) + (X^2 - X) + (X^3 - X^2) + 1)(X^2 + X + 1) \end{aligned}$$

Hence,

$$\Phi_{15}(X) = X^8 - X^7 + X^5 - X^4 + X^3 - X + 1$$

Question 2

Write a $\sqrt{7}$ in terms of roots of unity.

Solution. Let ζ be a 7-th root of unity. We calculate the map $a \mapsto a^{(7-1)/2}$ as

$$(1,2,3,4,5,6) \mapsto (1,1,-1,1,-1,-1)$$

Thus, we have seen that

$$\tau = \zeta + \zeta^2 - \zeta^3 + \zeta^4 - \zeta^5 - \zeta^6$$

satisfies $\tau^2 = (-1)^{(7-1)/2} \cdot 7 = -7$.

Now, $\iota^2 = -1$ is also a root of unity. Thus

$$\sqrt{7} = \iota(\zeta + \zeta^2 - \zeta^3 + \zeta^4 - \zeta^5 - \zeta^6)$$

is an expression in terms of roots of unity.

Note also that $\eta = \iota \cdot \zeta$ is a 28-th root of unity. So we can directly write

$$\sqrt{7} = \eta + \eta^9 + \eta^3 - \eta^{11} - \eta^5 - \eta^{13}$$